

SWC file for neuron's morphology, synapses, and compartmental split

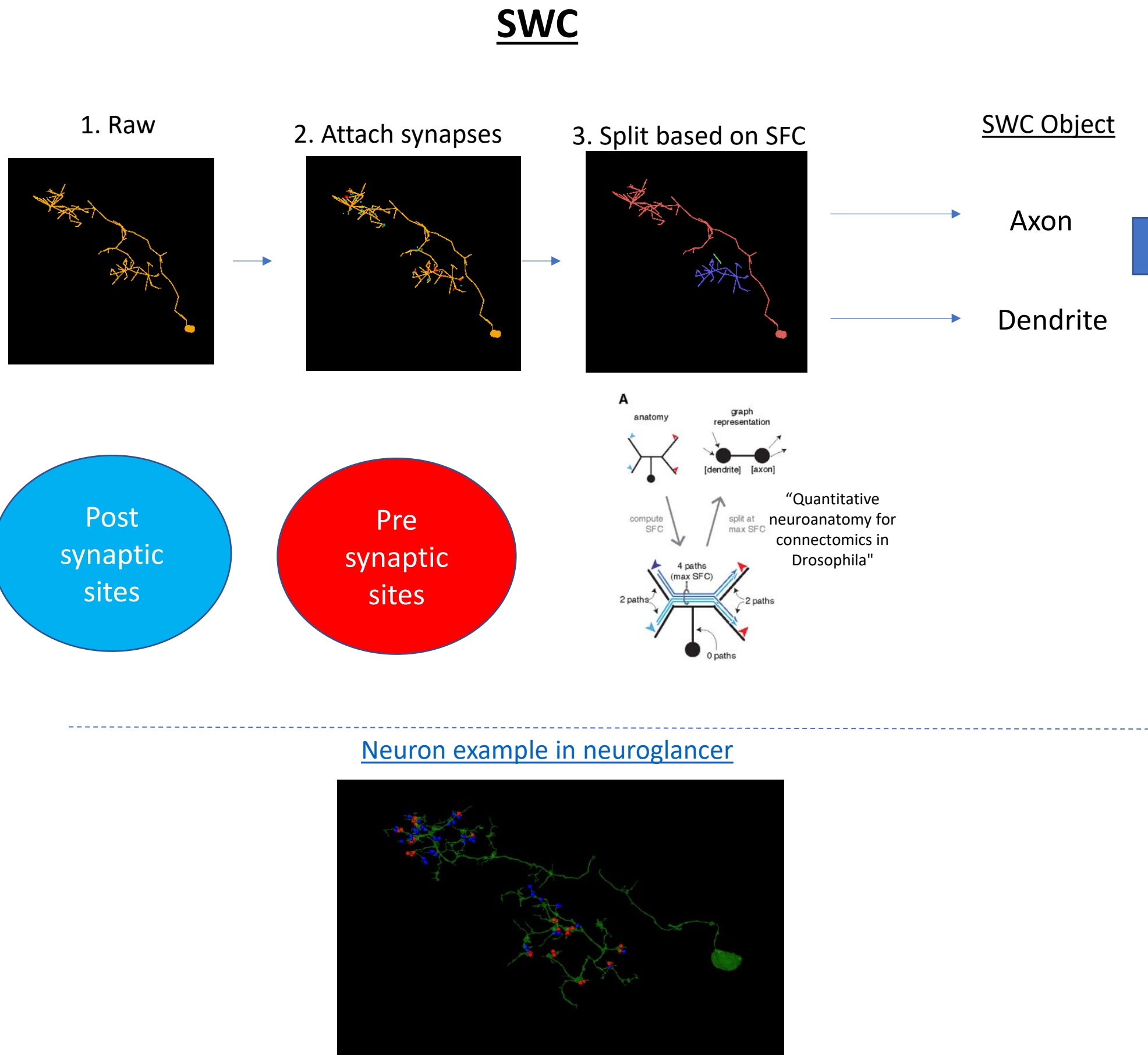
1. The SWC object contains spatial information about each neuron.
The geometric structure is embedded as a tree graph (therefore directed). Each node has the following information:

- Node ID
- Parent node ID
- X,Y,Z coordinates (The Full Adult Fly Brain “FAFB” project)
- Radius

*nodes were created to support the construction of the biological structure, therefore they have no actual biological meaning, such that twice as much nodes could have been created to build the same tree with twice as much nodes that are twice as much denser.

2. A table with all the synapses in the fly brain is used to attach to the swc all his synaptic sites (pre and post sites)

3. Using Synapse Flow Centrality (SFC) for splitting a neuron into two compartments based on pre and post synaptic sites distribution along the neural skeleton.

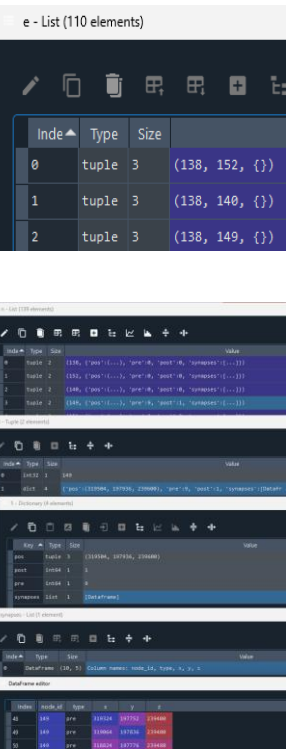


Generating a NetworkX Graph

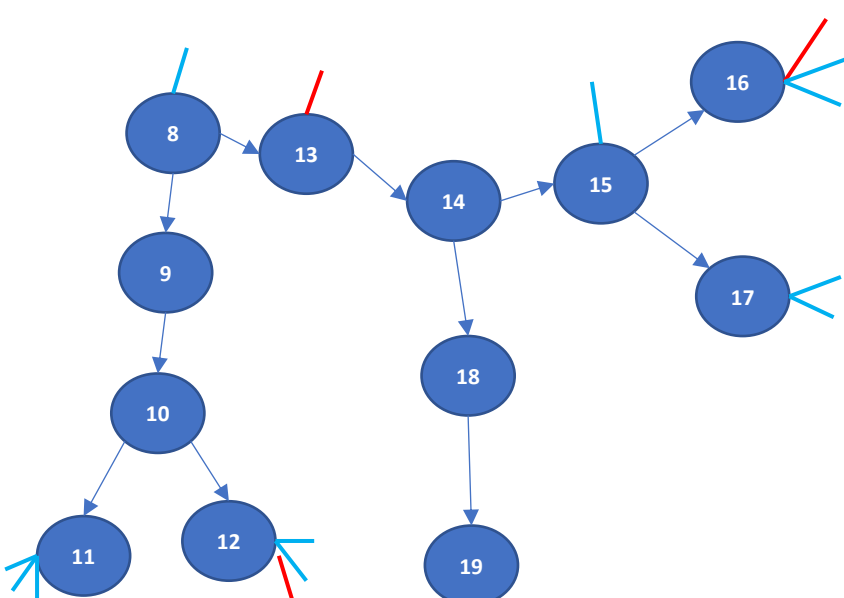
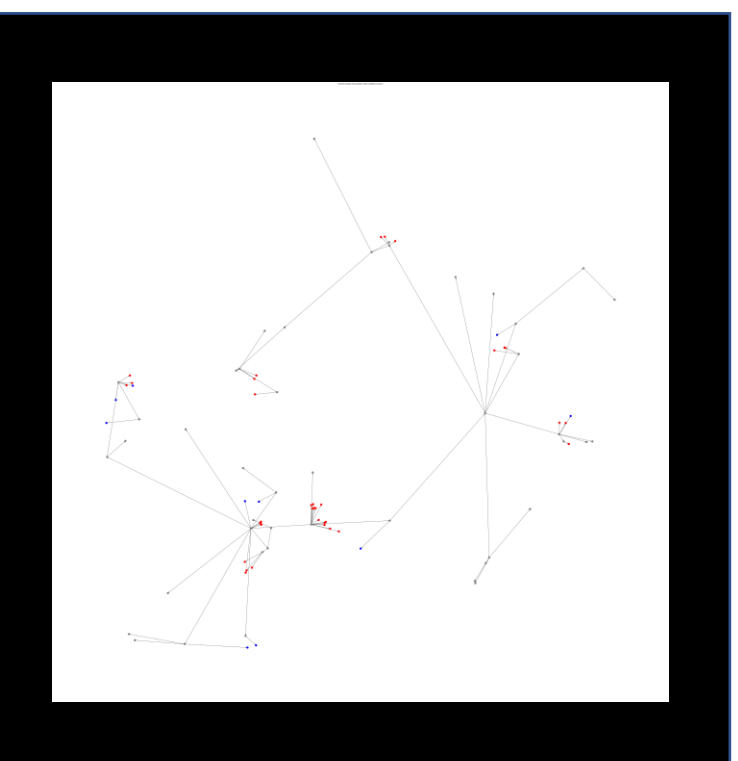
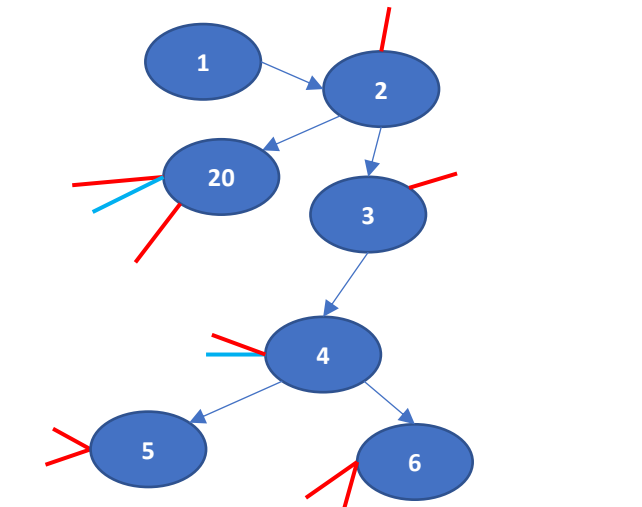
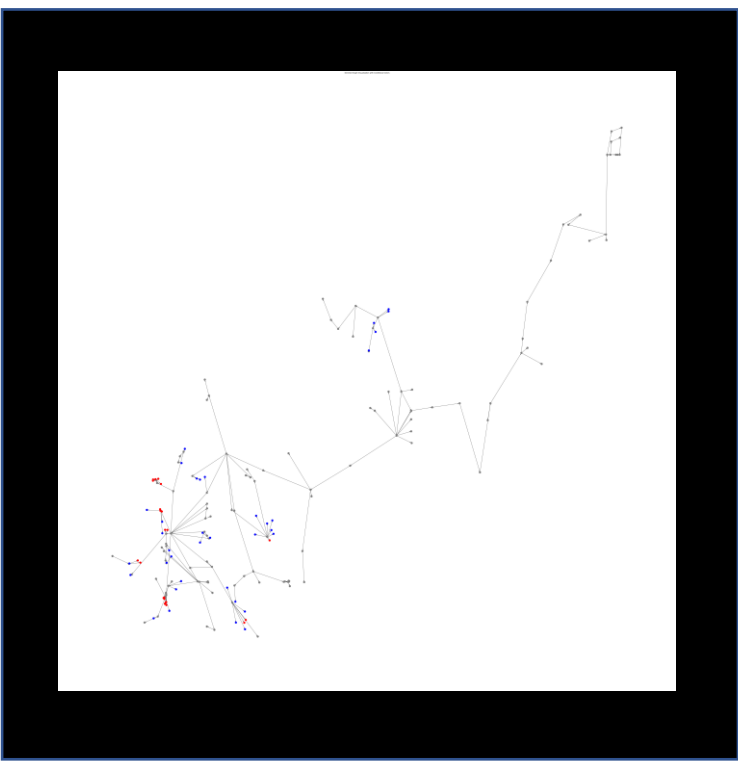
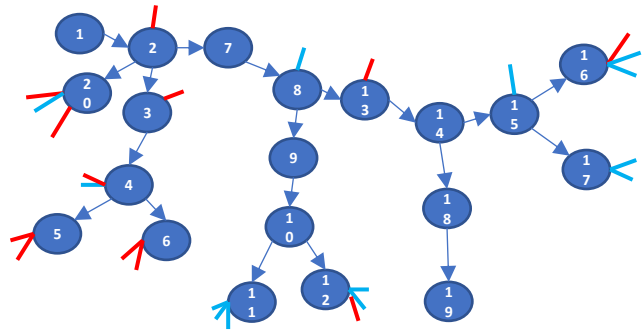
Based on the resulted split point, 2 new trees are created (axonic tree and dendritic tree) using networkX . The root of each tree is set to be the node that was connected to the split point.

Each networkx tree graph has the following info:

- Edges:**
- (Parent,Child)
- Nodes:**
- ID
 - XYZ coordinate
 - Synaptic sites :
 - Type (pre/post)
 - XYZ coordinates



Full Tree

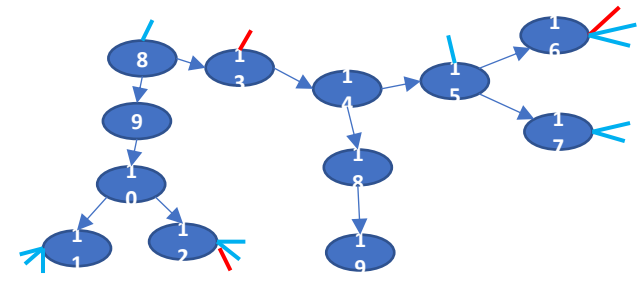


Pipeline

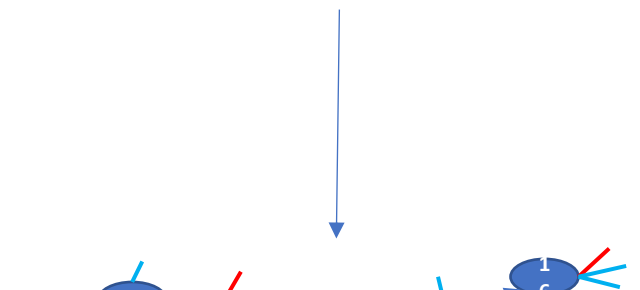
Graph compression

In these several steps, the original Tree-graph structure is compressed to a smaller tree based on predefined set of rules

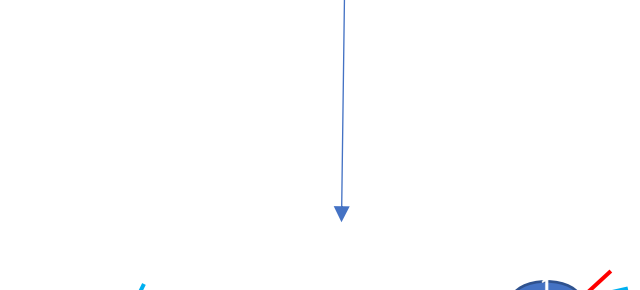
A dendrite or axon in first step



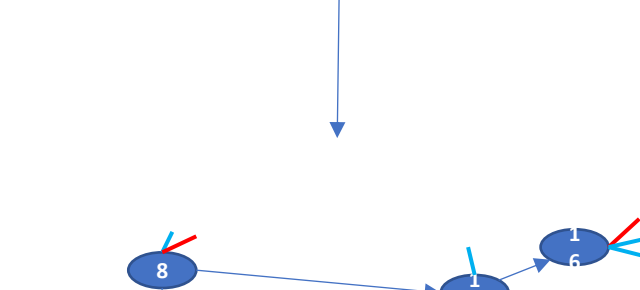
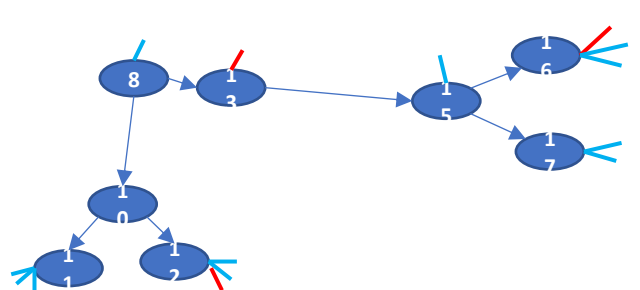
Prune leafs without synapses



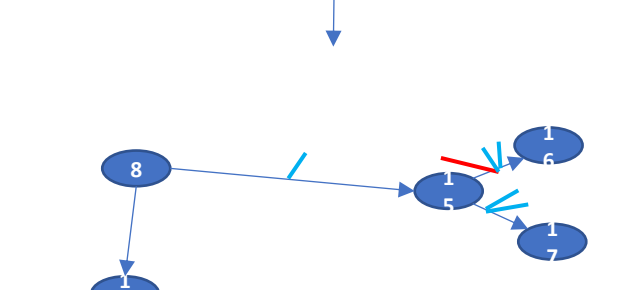
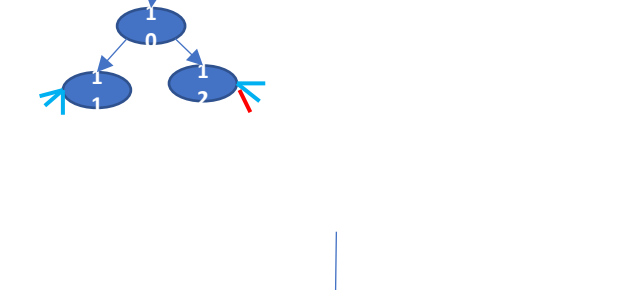
Prune nodes without synapses



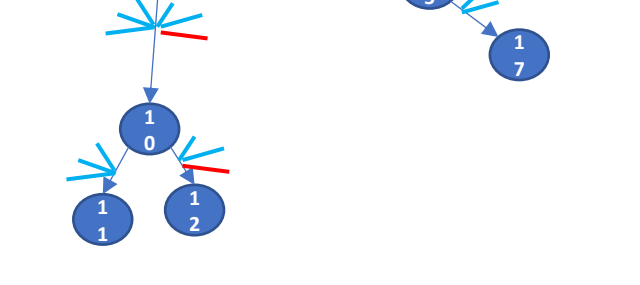
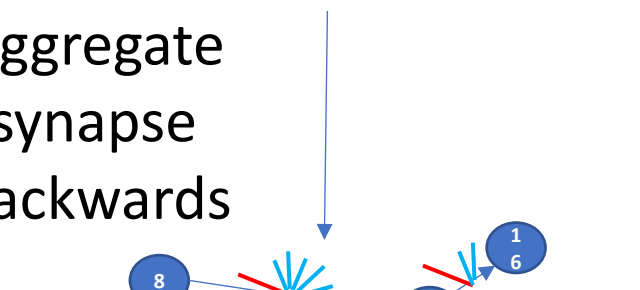
Compress tree such that each node is a leaf, or has more than one child



Move synapse to edges



This step is only relevant for the calculation step, since each node takes into account the entropy of pre/post synapses of his different Childs

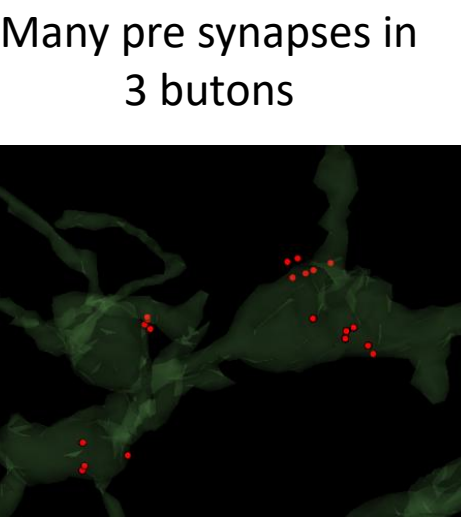
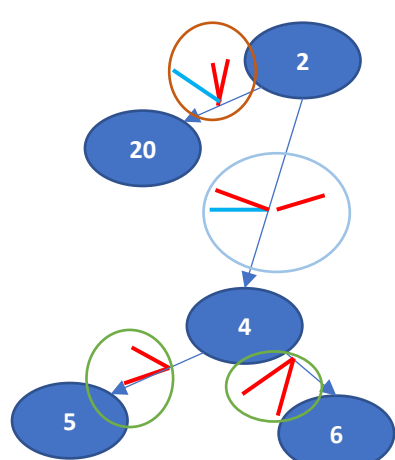


Node filtration

This step selects specific nodes to be ignored for future calculations based on the following criteria:

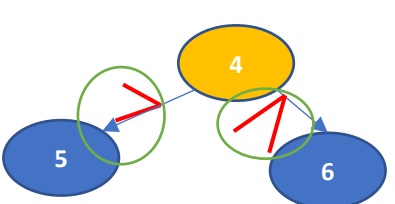
Pre synaptic sites

Aggr clustering pre synaptic sites with threshold distance of 1750nm



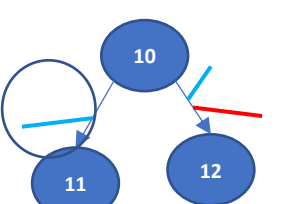
Clusutered tree in ng

If a node (node number 4 in this example) has pre synaptic sites from the same cluster with different children -> ignore him in the analysis pipeline

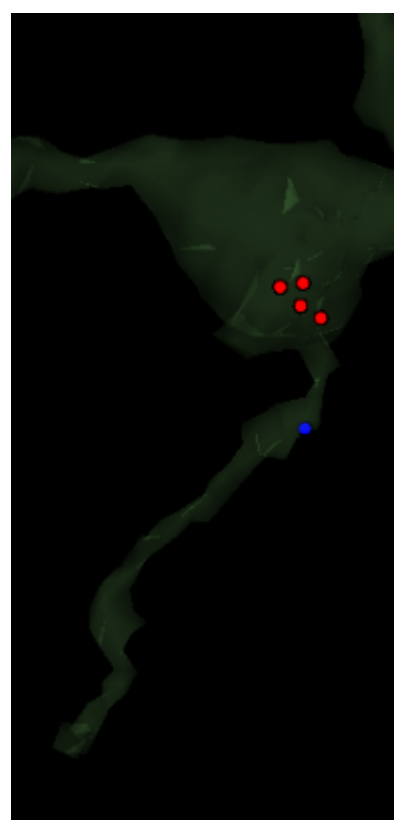


Post synaptic sites

If a connection between child and parent has only one post – don't use this connection for calculation in the analysis step



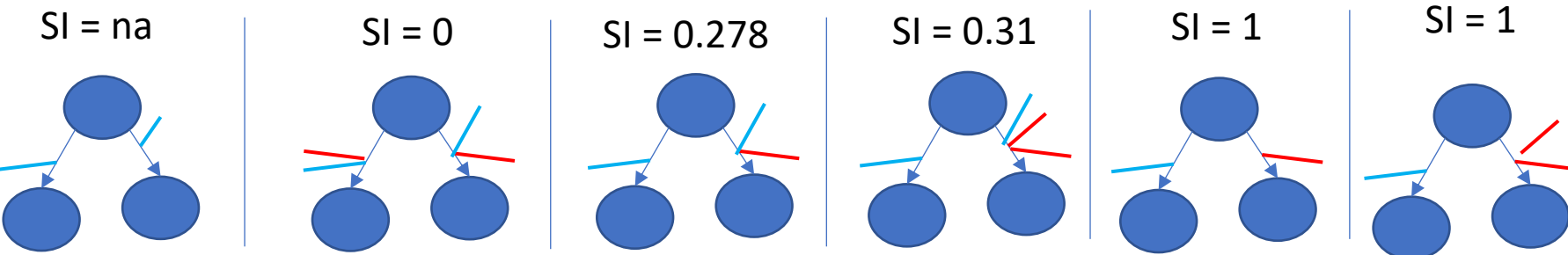
Example for one post synapse branch



Entropy calculation using SI

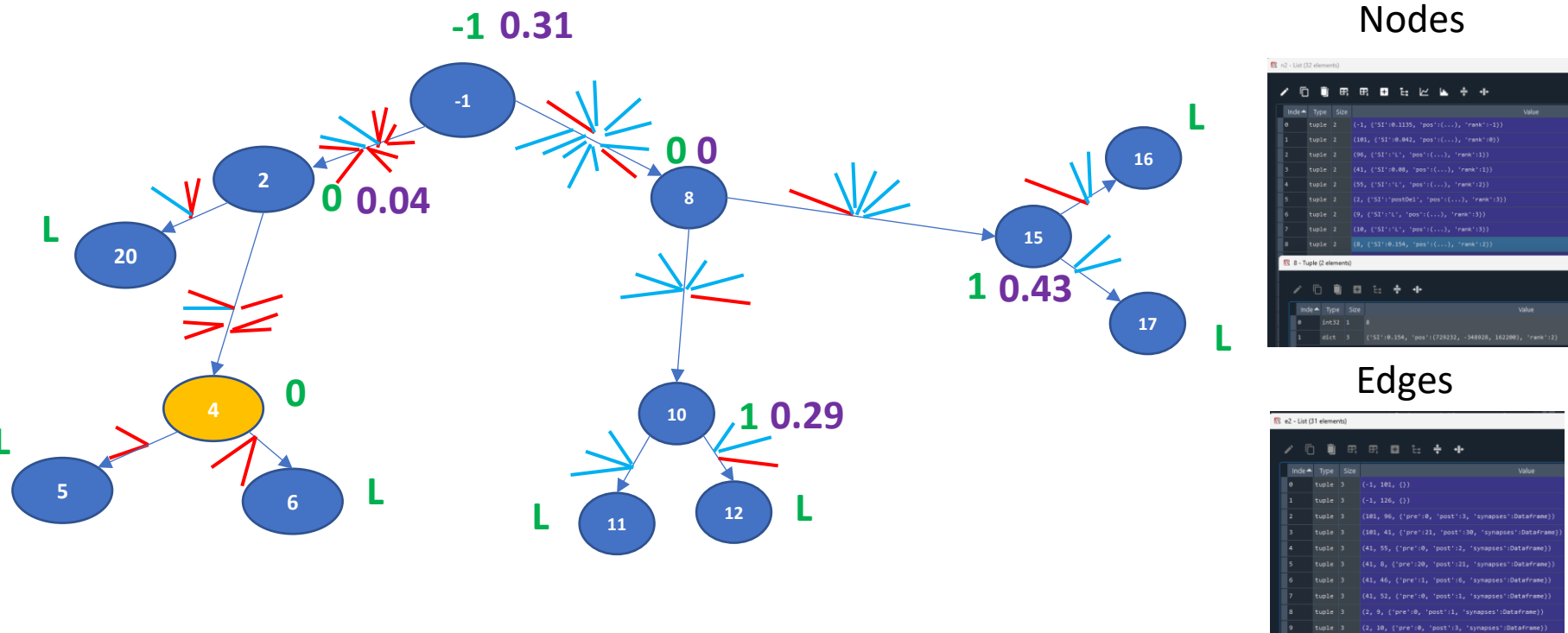
Leafs and nodes that were seleceted in the post processing will be considered in this step and will not receive a SI value

SI calculation



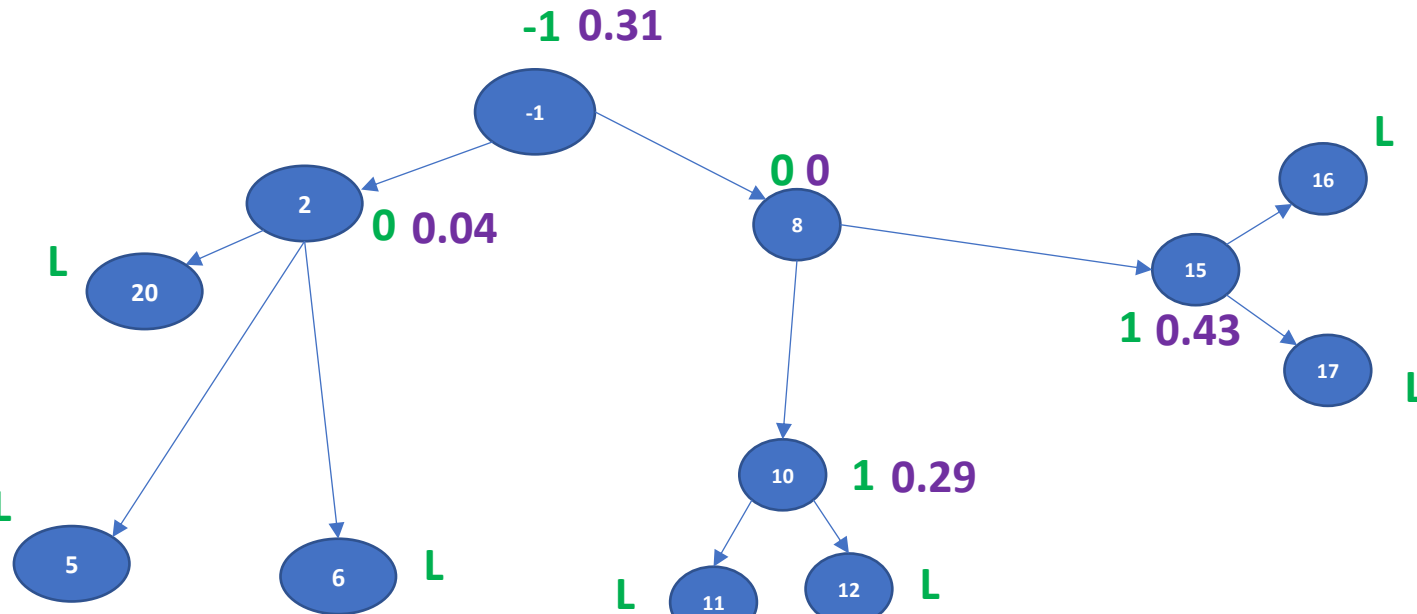
Full Tree with ranks

Depth SI



Final remodel

If a node doesn't have an SI, connect its parent and childs



Full Annotated Tree in ng

